MIL-C-51264A
21 June 1983
SUPERSEDING
MIL-C-51264
19 April 1968

MILITARY SPECIFICATION

CUPROUS CYANIDE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of cuprous cyanide.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging Of
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing - Interior Packaging

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical characteristics. Cuprous cyanide shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Copper (Cu)	70.0	----	4.2.4.1
Cyanide (CN)	28.5	----	4.2.4.2
Insoluble matter in NaCN solution	----	0.20	4.2.4.3

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the cuprous cyanide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For cuprous cyanide test. See 6.5 for sampling and testing precautions. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 100 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for cuprous cyanide test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Container liner missing or not as specified
(f) Unitization not as specified
(g) Marking incorrect, missing, or illegible

4.2.3.2 For cuprous cyanide test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one unit container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.0 percent defective.

4.2.4 Cuprous cyanide tests. See 6.5 for sampling and testing precautions. Water in accordance with ASTM D1193, Type II and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 Copper. Weigh to the nearest 0.1 milligram (mg) approximately 0.3 g of the specimen into a 250-milliliter (mL) Erlenmeyer flask. Add 10 mL of 1 part nitric acid to 3 parts water in a hood. Heat until all the hydrogen cyanide and oxides of nitrogen are removed. Add 5 mL of saturated bromine water and heat until fumes of bromine cease. Dilute to 25 mL with cold water and neutralize with concentrated ammonium hydroxide solution, drop by drop, until a blue color just appears. Add 5 mL of concentrated acetic acid, cool to 10° ± 2°C, add 1 crystal of potassium fluoride and 20 g of potassium iodide, and mix thoroughly. Titrate the liberated iodine with sodium thiosulfate solution which has been standardized until the brownish tint or yellow color has nearly disappeared, then add 5 mL of starch solution and continue the titration until the solution changes from the blue to yellowish-white. Calculate the percent by weight copper as follows:

$$\text{Percent copper} = \frac{6.36\ AB}{W}$$

where: A = Milliliters of sodium thiosulfate solution used,
B = Normality of sodium thiosulfate solution, and
W = Weight of specimen, in grams.

4.2.4.2 Cyanide (CN). Pipet 10 mL of sodium cyanide solution (made alkaline with 10 g of sodium hydroxide per liter) into each of 2 Erlenmeyer flasks. Weigh to the nearest milligram approximately 0.25 g of the specimen, transfer to one of the flasks, and dissolve. Add 5 mL of 10-percent potassium iodide solution to each flask and titrate both solutions with 0.1N silver nitrate solution to a faint, permanent turbidity. Calculate the percent by weight cyanide as follows:

$$\text{Percent cyanide} = \frac{2.60\ A(B - C)}{W}$$

where: A = Normality of silver nitrate solution,
B = Milliliters of silver nitrate solution used to titrate flask without specimen,
C = Milliliters of silver nitrate solution used to titrate flask with specimen, and
W = Weight of specimen, in grams.

4.2.4.3 Insoluble matter in sodium cyanide solution. Dissolve 5.8 g of sodium cyanide in 50 mL of water at 50° ± 2°C. Weigh to the nearest milligram approximately 5.5 g of the specimen. Add the specimen slowly to the sodium cyanide solution with stirring until it is completely dissolved. Dilute to 200 mL with water, filter through a tared filter crucible, and wash with water. Dry to constant weight at 110° ± 2°C, cool to room temperature, and weigh. Calculate the percent by weight insoluble matter in sodium cyanide solution as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A - B)}{W}$$

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where: A = Weight of crucible and residue, in grams,
B = Weight of crucible, in grams, and
W = Weight of specimen, in grams.

4.2.5 Container leakage test. Hold the container in each of the following positions and moderately shake for a minimum period of 1 minute:

(a) Upright
(b) Upside down
(c) On one side
(d) On other side

Examine the container after each period for any evidence of leakage.

5. PACKAGING

5.1 Unit packing. Cuprous cyanide shall be unit packed level A or C as specified (see 6.2).

5.1.1 Level A. A 5-pound (lb) quantity (+0 or -0.75 ounce) of cuprous cyanide shall be unit packed in a can in accordance with the applicable requirements of PPP-C-2020. The can shall conform to type V, class 1 or 2 round of PPP-C-96, using No. 100 tinplate minimum, and exterior coating plan B. The can shall be furnished with a polyethylene bag liner conforming to type I, class B, style 2 of MIL-B-117, closed by heat sealing, tying or knotting. The can shall be securely closed, preventing loosening in storage and transit. There shall be no evidence of leakage when tested as specified in 4.2.5.

5.1.2 Level C. A quantity of 5 lb (+0 or -0.75 ounce) of cuprous cyanide shall be unit packed level C in a manner to protect the specified purity of the cuprous cyanide, and the integrity of the container from the supply source to the first destination, and for a minimum duration of six months.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Six cans of cuprous cyanide, unit packed as specified in 5.1.1, shall be packed for shipment in a nailed wood box conforming to class 2, style 4, for a type 2 average load of PPP-B-621. Fiberboard separators, type CF, class weather-resistant, variety SW, grade W5c of PPP-F-320 shall be placed between unit packs where needed to prevent motion of contents. The box shall be closed and strapped in accordance with the applicable requirements as specified in the appendix to PPP-B-621.

5.2.2 Level B. Cuprous cyanide shall be packed level B as specified in 5.2.1, except that a fiberboard box shall be used instead of the nailed wood box. The fiberboard box shall conform to RSC, grade V3c of PPP-B-636 and shall be closed by taping in accordance with the requirements for taping style RSC boxes to be shipped as exterior containers, and then reinforced by tape banding as specified in the appendix to PPP-B-636.

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5.2.3 Level C. Cuprous cyanide, unit packed as specified in 5.1, shall be packed in accordance with Department of Transportation (DOT) and any other applicable regulations in a manner to assure acceptance by the carrier and safe delivery to the first destination. Containers shall be in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, and the rules applicable to any other selected mode of transporation.

5.3 Unitization. Level A shipping containers shall be palletized in accordance with load type I using the fiberboard cap of MIL-STD-147. The pallet shall conform to type IV of NN-P-71. Level C shipping containers shall be unitized in a manner to assure acceptance by the carrier, safe delivery to first destination, and stable stacking in storage.

5.4 Marking. Each container and unitized load shall be marked in accordance with MIL-STD-129, DOT regulations, and any other applicable regulations. Each container shall be marked to show lot or batch number and date of manufacture of the cuprous cyanide. In addition, each unit container shall be marked with the following precautionary marking:

POISON

Irritating to eyes, skin, nose and throat.

If inhaled or swallowed will cause dizziness or loss of consciousness.

FIRST AID:

SEEK MEDICAL ATTENTION IMMEDIATELY IN ALL CASES

Remove contaminated clothing.

Flush affected areas with copious amounts of water.

If in eyes, flush with steady flow of water for 15 minutes.

If swallowed and victim is conscious, have victim induce vomiting by touching back of throat with finger.

If swallowed and victim is unconscious or having convulsions, keep victim warm.

If breathing has stopped, give artificial respiration.

If breathing is difficult, give oxygen.

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6. NOTES

6.1 Intended use. Cuprous cyanide is intended for use in preparing copper and brass cyanide electroplating solutions.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification and the level of unit packing and packing required (see 5.1 and 5.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially hazardous to personnel. Tests should be performed under an exhaust hood due to the possibility of toxic gases being liberated during testing. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material.

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-B385

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-51264A	CUPROUS CYANIDE, TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE SUBMISSION *(YYMMDD)*

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